

Tutorial 2 – Solutions

ECON 101 – Macroeconomics

Fall 2025

Problem 1: Expenditure Approach

Given

All values are in billions of dollars.

$$C = 1,800, \quad I = 600, \quad G = 720, \quad X = 450, \quad M = 520.$$

Key definitions

GDP (expenditure approach) adds spending on final goods and services:

$$Y = C + I + G + (X - M) \equiv C + I + G + \text{NX},$$

where $\text{NX} = X - M$ is net exports.

GDP per capita is GDP divided by population.

Output gap (%) is $\frac{Y - Y^*}{Y^*} \times 100$, where Y is actual GDP and Y^* is potential GDP.

(a) Nominal GDP

First compute net exports:

$$\text{NX} = X - M = 450 - 520 = -70.$$

Then GDP:

$$Y = 1,800 + 600 + 720 + (-70) = 3,050.$$

Answer: \$3,050 billion.

(b) GDP per capita

Population = 40 million. Convert units consistently:

$$\text{GDP per capita} = \frac{3,050 \text{ billion}}{40 \text{ million}} = \frac{3,050 \times 10^9}{40 \times 10^6} = 76,250.$$

Answer: \$76,250 per person.

(c) Output gap

Potential GDP $Y^* = 3,800$ billion.

$$\text{Output gap} = \frac{Y - Y^*}{Y^*} \times 100 = \frac{3,050 - 3,800}{3,800} \times 100 = \frac{-750}{3,800} \times 100 \approx -19.74\%.$$

Answer: $\boxed{-19.7\% \text{ (recessionary gap)}}$.

Interpretation. A negative gap means the economy is producing below its sustainable capacity.

Problem 2: Income Approach

Given

All values are in billions.

Wages = 1,500, Rent = 200, Interest = 250, Profits = 400, Indirect taxes (less subsidies) = 250, Depreciation = 300

Key definitions

Net Domestic Income (factor cost) sums factor payments:

$$\text{NDI}_{\text{fc}} = \text{Wages} + \text{Rent} + \text{Interest} + \text{Profits}.$$

GDP at market prices adds indirect taxes and depreciation to net domestic income:

$$\text{GDP}_{\text{mp}} = \text{NDI}_{\text{fc}} + \text{Indirect taxes (less subsidies)} + \text{Depreciation}.$$

(a) Net Domestic Income at factor cost

$$\text{NDI}_{\text{fc}} = 1,500 + 200 + 250 + 400 = 2,350.$$

Answer: $\boxed{\$2,350 \text{ billion}}$.

(b) GDP at market prices

$$\text{GDP}_{\text{mp}} = 2,350 + 250 + 300 = 2,900.$$

Answer: $\boxed{\$2,900 \text{ billion}}$.

(c) Comparison to Problem 1

Problem 1 found $Y = 3,050$ (expenditure side). Here we have 2,900 (income side). The difference is

$$\text{Statistical discrepancy} = 3,050 - 2,900 = 150 \text{ billion}.$$

Interpretation. Conceptually, expenditure and income measures of GDP are equal, but in practice data come from different sources and timing, so statistical discrepancies arise.

Problem 3: Nominal vs. Real GDP and the GDP Deflator

Data

Year	Qty Apples	Price Apples	Qty Laptops	Price Laptops
2022	1,000	\$2	200	\$1,000
2023	1,200	\$3	220	\$1,100
2024	1,300	\$3	250	\$1,200

Key definitions

Nominal GDP uses current-year prices and quantities.

Real GDP (base 2022) values quantities at 2022 prices.

GDP deflator = $\frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$. It is a broad price index for domestically produced final goods and services.

(a) Nominal GDP

Compute each year:

$$\text{Nominal 2022} = 1,000 \cdot 2 + 200 \cdot 1,000 = 2,000 + 200,000 = \$202,000,$$

$$\text{Nominal 2023} = 1,200 \cdot 3 + 220 \cdot 1,100 = 3,600 + 242,000 = \$245,600,$$

$$\text{Nominal 2024} = 1,300 \cdot 3 + 250 \cdot 1,200 = 3,900 + 300,000 = \$303,900.$$

(b) Real GDP (base 2022)

Use 2022 prices: $p_A = 2$, $p_L = 1,000$.

$$\text{Real 2022} = 1,000 \cdot 2 + 200 \cdot 1,000 = \$202,000,$$

$$\text{Real 2023} = 1,200 \cdot 2 + 220 \cdot 1,000 = 2,400 + 220,000 = \$222,400,$$

$$\text{Real 2024} = 1,300 \cdot 2 + 250 \cdot 1,000 = 2,600 + 250,000 = \$252,600.$$

(c) GDP deflator

$$\text{Deflator 2022} = \frac{202,000}{202,000} \times 100 = 100.0,$$

$$\text{Deflator 2023} = \frac{245,600}{222,400} \times 100 \approx 110.4,$$

$$\text{Deflator 2024} = \frac{303,900}{252,600} \times 100 \approx 120.37.$$

Interpretation. The price level of domestically produced final goods and services rose about 10.4% from 2022 to 2023 and about 9.0% from 2023 to 2024 (see next part).

(d) Inflation rate (GDP deflator) from 2023 to 2024

$$\pi_{23 \rightarrow 24} = \frac{120.37 - 110.4}{110.4} \times 100 \approx 9.03\%.$$

Answer: 9.0% (approximately).

Problem 4: Underground Economy Adjustment

Given

Measured GDP = \$2,900 billion. Underground economy = 7% of measured GDP. Population = 39 million.

(a) Size of underground economy

$$0.07 \times 2,900 = 203.$$

Answer: \$203 billion.

(b) Adjusted GDP including underground activity

$$\text{Adjusted GDP} = 2,900 + 203 = \text{3,103 billion dollars}.$$

(c) Effect on GDP per capita

Measured GDP per capita:

$$\frac{2,900 \times 10^9}{39 \times 10^6} = 74,358.97 \approx \$74,359.$$

Adjusted GDP per capita:

$$\frac{3,103 \times 10^9}{39 \times 10^6} = 79,564.10 \approx \$79,564.$$

Difference:

$$79,564 - 74,359 = \text{5,205 per person (approx.)}.$$

(d) Why exclusion distorts comparisons

Countries differ in how much activity is unreported and in enforcement quality. If Country A has more unreported activity than Country B, measured GDP understates A relative to B, biasing international comparisons of living standards and productivity.

Problem 5: Chain-Weighted Real GDP

Data

Year	Qty Coffee	Price Coffee	Qty Textbooks	Price Textbooks
2023	500	\$4	100	\$60
2024	600	\$5	120	\$70

Key definitions

Nominal GDP in year t : $\sum_i p_{it}q_{it}$.

Laspeyres real GDP growth uses base-year prices.

Paasche real GDP growth uses current-year prices.

Chain-weighted growth between $t - 1$ and t :

$$1 + g_{\text{chain}} = \sqrt{(1 + g_L)(1 + g_P)} \implies g_{\text{chain}} = \sqrt{(1 + g_L)(1 + g_P)} - 1.$$

(a) Nominal GDP

$$\text{Nominal 2023} = 500 \cdot 4 + 100 \cdot 60 = 2,000 + 6,000 = \$8,000,$$

$$\text{Nominal 2024} = 600 \cdot 5 + 120 \cdot 70 = 3,000 + 8,400 = \$11,400.$$

(b) Real 2024 using 2023 prices (Laspeyres numerator)

Use 2023 prices: coffee \$4, textbook \$60.

$$\text{Real 2024@2023}p = 600 \cdot 4 + 120 \cdot 60 = 2,400 + 7,200 = \boxed{\$9,600}.$$

(c) Real 2023 using 2024 prices (Paasche denominator)

Use 2024 prices: coffee \$5, textbook \$70.

$$\text{Real 2023@2024}p = 500 \cdot 5 + 100 \cdot 70 = 2,500 + 7,000 = \boxed{\$9,500}.$$

(d) Chain-weighted real GDP growth from 2023 to 2024

Laspeyres growth (base 2023 prices):

$$g_L = \frac{\text{Real 2024@2023}p - \text{Real 2023@2023}p}{\text{Real 2023@2023}p} = \frac{9,600 - 8,000}{8,000} = 0.20.$$

Paasche growth (current 2024 prices):

$$g_P = \frac{\text{Real 2024@2024}p - \text{Real 2023@2024}p}{\text{Real 2023@2024}p} = \frac{11,400 - 9,500}{9,500} = 0.20.$$

Chain-weighted growth:

$$1 + g_{\text{chain}} = \sqrt{(1 + 0.20)(1 + 0.20)} = \sqrt{1.2 \cdot 1.2} = 1.2 \implies g_{\text{chain}} = 0.20.$$

Answer: $\boxed{20\%}$.

(e) Why chain-weighting is preferred

It updates expenditure weights each period and takes the geometric average of Laspeyres and Paasche growth rates. This reduces substitution bias that occurs when consumers shift toward relatively cheaper goods, something a fixed-base index can miss.

Problem 6: GDP, Hours, and Productivity

Given

Both countries have GDP per capita = \$40,000. Hours per worker:

Country A: 2,200 hours/year, Country B: 1,600 hours/year.

Key definitions

Labour productivity per hour $\approx \frac{\text{GDP per worker}}{\text{hours per worker}}$. With only GDP per capita given, assume, for illustration, similar population-to-worker ratios so that comparisons based on GDP per capita per hour are informative.

(a) GDP per hour worked

$$\text{A: } \frac{40,000}{2,200} \approx \$18.18 \text{ per hour,}$$

$$\text{B: } \frac{40,000}{1,600} = \$25.00 \text{ per hour.}$$

Answer: A $\approx$ \$18.18/hr, B = \$25.00/hr.

(b) Which country is more productive?

Country B, because it produces the same GDP per person with fewer hours, implying higher output per hour.

(c) Why GDP per capita alone is incomplete

GDP per capita ignores hours worked, leisure, household production, income distribution, environmental quality, and non-market benefits. A country can have the same GDP per capita with fewer hours, indicating higher productivity and possibly higher well-being due to more leisure.

Appendix: Quick Reference Formulas

- Expenditure GDP: $Y = C + I + G + (X - M)$.
- Income GDP: $\text{GDP}_{\text{mp}} = \text{Wages} + \text{Rent} + \text{Interest} + \text{Profits} + \text{Indirect taxes (less subsidies)} + \text{Depreciation}$.
- Nominal GDP: $\sum_i p_{it}q_{it}$. Real GDP (base b): $\sum_i p_{ib}q_{it}$.
- GDP deflator: $\frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$.
- Inflation (deflator): $\frac{P_t - P_{t-1}}{P_{t-1}} \times 100$.
- Chain growth: $1 + g_{\text{chain}} = \sqrt{(1 + g_L)(1 + g_P)}$.